

# Probabilistic Graphical Models

## CSE 5830

Representation

Lecture 6

Markov Network and  
Conditional Random Fields  
(CRF)

# Goals for Today

- MN factorization and independencies
- Log-linear Markov networks
- Conditional random fields

# Active Trails (revisited)

- A trail  $X_1 - \dots - X_n$  is active given  $Z$  if no  $X_i$  is in  $Z$
- Active trails depend only on graph structure
- We can talk about flow of influence to define independency
- There are different types of independency
  - Global Markov dependency

$$\mathcal{I}(\mathcal{H}) = \{(X \perp Y \mid Z) : \text{sep}_{\mathcal{H}}(X; Y \mid Z)\}$$

- Pairwise Markov dependency

$$\mathcal{I}_p(\mathcal{H}) = \{(X \perp Y \mid \mathcal{X} - \{X, Y\}) : X - Y \notin \mathcal{H}\}$$

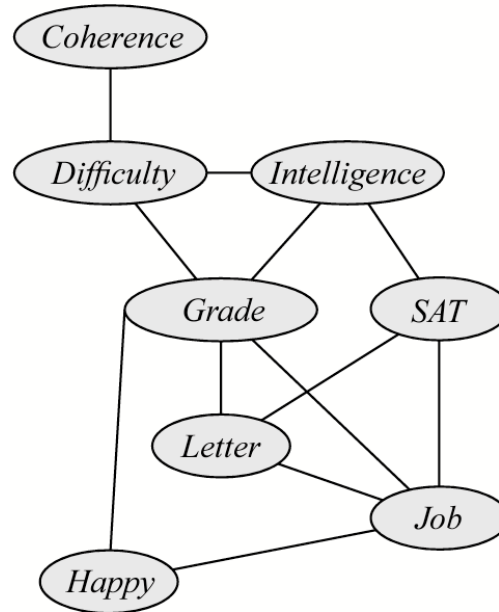
- Local Markov dependency

$$\mathcal{I}_\ell(\mathcal{H}) = \{(X \perp \mathcal{X} - \{X\} - \text{MB}_{\mathcal{H}}(X) \mid \text{MB}_{\mathcal{H}}(X)) : X \in \mathcal{X}\}$$

# Quiz

Consider the graph and the given independencies below. Which of the following is not correct?

- a)  $I \perp J | G, S$
- b)  $C \perp J | S, L$
- c)  $C \perp H | I, G$
- d)  $I \perp H | G, J$



## PGM Quiz 2

<https://forms.cloud.microsoft/r/RzPejGZSQr>



# Independency and MNs Representation (revisited)

Markov networks ...

The Markov network's  
graph is an I-map for P.



$$P(\mathbf{X}) = \frac{1}{Z} \prod_{i=1}^m \phi_i(\mathbf{D}_i)$$

# Using tables to represent probability distributions

Example:

$\phi_1(A, B)$

|       |       |    |
|-------|-------|----|
| $a^0$ | $b^0$ | 30 |
| $a^0$ | $b^1$ | 5  |
| $a^1$ | $b^0$ | 1  |
| $a^1$ | $b^1$ | 10 |

$\phi_4(D, A)$

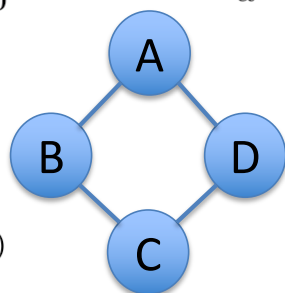
|       |       |     |
|-------|-------|-----|
| $d^0$ | $a^0$ | 100 |
| $d^0$ | $a^1$ | 1   |
| $d^1$ | $a^0$ | 1   |
| $d^1$ | $a^1$ | 100 |

$\phi_2(B, C)$

|       |       |     |
|-------|-------|-----|
| $b^0$ | $c^0$ | 100 |
| $b^0$ | $c^1$ | 1   |
| $b^1$ | $c^0$ | 1   |
| $b^1$ | $c^1$ | 100 |

$\phi_3(C, D)$

|       |       |     |
|-------|-------|-----|
| $c^0$ | $d^0$ | 1   |
| $c^0$ | $d^1$ | 100 |
| $c^1$ | $d^0$ | 100 |
| $c^1$ | $d^1$ | 1   |



| Assignment |       |       |       | Unnormalized | Normalized          |
|------------|-------|-------|-------|--------------|---------------------|
| $a^0$      | $b^0$ | $c^0$ | $d^0$ | 300,000      | 0.04                |
| $a^0$      | $b^0$ | $c^0$ | $d^1$ | 300,000      | 0.04                |
| $a^0$      | $b^0$ | $c^1$ | $d^0$ | 300,000      | 0.04                |
| $a^0$      | $b^0$ | $c^1$ | $d^1$ | 30           | $4.1 \cdot 10^{-6}$ |
| $a^0$      | $b^1$ | $c^0$ | $d^0$ | 500          | $6.9 \cdot 10^{-5}$ |
| $a^0$      | $b^1$ | $c^0$ | $d^1$ | 500          | $6.9 \cdot 10^{-5}$ |
| $a^0$      | $b^1$ | $c^1$ | $d^0$ | 5,000,000    | 0.69                |
| $a^0$      | $b^1$ | $c^1$ | $d^1$ | 500          | $6.9 \cdot 10^{-5}$ |
| $a^1$      | $b^0$ | $c^0$ | $d^0$ | 100          | $1.4 \cdot 10^{-5}$ |
| $a^1$      | $b^0$ | $c^0$ | $d^1$ | 1,000,000    | 0.14                |
| $a^1$      | $b^0$ | $c^1$ | $d^0$ | 100          | $1.4 \cdot 10^{-5}$ |
| $a^1$      | $b^0$ | $c^1$ | $d^1$ | 100          | $1.4 \cdot 10^{-5}$ |
| $a^1$      | $b^1$ | $c^0$ | $d^0$ | 10           | $1.4 \cdot 10^{-6}$ |
| $a^1$      | $b^1$ | $c^0$ | $d^1$ | 100,000      | 0.014               |
| $a^1$      | $b^1$ | $c^1$ | $d^0$ | 100,000      | 0.014               |
| $a^1$      | $b^1$ | $c^1$ | $d^1$ | 100,000      | 0.014               |

# Alternative parameterization

- General representation
  - Using local structure (not tables) to represent probability distributions
  - Let  $\mathbf{D}_i$  denote the set of variables (subset of  $\mathbf{X}$ ) in the  $i^{th}$  complete sub graph (clique).
  - Probability distribution is a Gibbs distribution:

$$\begin{aligned}P(\mathbf{X}) &= \frac{U(\mathbf{X})}{Z} \\U(\mathbf{X}) &= \prod_{i=1}^m \phi_i(\mathbf{D}_i) \\Z &= \sum_{\mathbf{x} \in \text{Val}(\mathbf{X})} U(\mathbf{x})\end{aligned}$$

# Alternative parameterization

- Logarithmic Representation

- Logarithmic:  $\phi(D_i) = e^{-\psi_i(D_i)}$

$$P(\mathbf{X}) = \frac{U(\mathbf{X})}{Z}$$

$$U(\mathbf{X}) = \prod_{i=1}^m \phi_i(\mathbf{D}_i)$$

$$P(\mathbf{X}) = \frac{1}{Z} e^{-\sum_i \psi_i(\mathbf{D}_i)}$$

$$\text{Energy (lower energy = higher probability)} = -\sum_i \psi_i(\mathbf{D}_i)$$

# Representing factors in exponential or logarithmic

- Using Features
- A feature is a function  $f : \text{Val}(\mathbf{D}_i) \rightarrow \mathbb{R}$ 
  - A factor without the positivity requirement
  - Example: indicator feature
    - Indicator function for some events
    - E.g. two super-pixels have the same color

$$\psi(\mathbf{D}_i) = \begin{cases} 1 & \text{for some } y \in \mathbf{D} \\ 0 & \text{for others} \end{cases}$$

$$\phi(D_i) = e^{-\psi_i(D_i)}$$

$$P(\mathbf{X}) = \frac{1}{Z} \prod_{i=1}^m \underbrace{\phi_i(D_i)}_{\text{factors}} = \frac{1}{Z} e^{-\sum_i \underbrace{\psi_i(D_i)}_{\text{features}}} \Rightarrow \ln[P(X)] = -\ln Z - \sum_{i=1} \psi_i(\mathbf{D}_i)$$

# MN representation

Three representation of parameterization of an MN

- Product of potential on cliques

- Product of factors

$$P(\mathbf{X}) = \frac{1}{Z} \prod_{i=1}^m \phi_i(\mathbf{D}_i)$$

- Sum of features for  $\ln P$

$$P(\mathbf{X}) = \frac{1}{Z} e^{-\sum_i \psi_i(\mathbf{D}_i)} \quad \Rightarrow \quad \ln[P(X)] = -\ln Z - \sum_{i=1} \psi_i(\mathbf{D}_i)$$

# Example

$$\phi_1(A, B) = \exp(-\psi_1(A, B))$$

$$\psi_1(A, B) = -\ln(\phi_1(A, B))$$

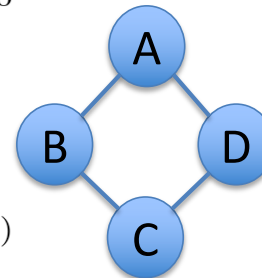
|       |       |       |
|-------|-------|-------|
| $a^0$ | $b^0$ | -3.4  |
| $a^0$ | $b^1$ | -1.61 |
| $a^1$ | $b^0$ | 0     |
| $a^1$ | $b^1$ | -2.3  |

$$\phi_1(A, B)$$

|       |       |    |
|-------|-------|----|
| $a^0$ | $b^0$ | 30 |
| $a^0$ | $b^1$ | 5  |
| $a^1$ | $b^0$ | 1  |
| $a^1$ | $b^1$ | 10 |

$$\phi_4(D, A)$$

|       |       |     |
|-------|-------|-----|
| $d^0$ | $a^0$ | 100 |
| $d^0$ | $a^1$ | 1   |
| $d^1$ | $a^0$ | 1   |
| $d^1$ | $a^1$ | 100 |



$$\phi_2(B, C)$$

|       |       |     |
|-------|-------|-----|
| $b^0$ | $c^0$ | 100 |
| $b^0$ | $c^1$ | 1   |
| $b^1$ | $c^0$ | 1   |
| $b^1$ | $c^1$ | 100 |

$$\phi_3(C, D)$$

|       |       |     |
|-------|-------|-----|
| $c^0$ | $d^0$ | 1   |
| $c^0$ | $d^1$ | 100 |
| $c^1$ | $d^0$ | 100 |
| $c^1$ | $d^1$ | 1   |

$$P(A, B, C, D) \propto \prod_{i=1,2,3,4} \exp(-\psi_1(D_i))$$

$$P(A, B, C, D) \propto \exp[-\psi_1(A, B) - \psi_2(B, C) - \psi_3(C, D) - \psi_4(D, A)]$$

# Log-linear Markov networks

- A set of features, where  $\mathbf{D}_i$  is a complete subgraph in  $H$

$$\mathcal{F} = \{f_1(\mathbf{D}_1), \dots, f_k(\mathbf{D}_k)\}$$

- Set of weights  $w_1, \dots, w_k,$

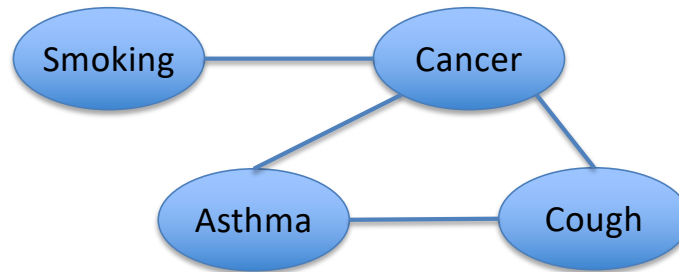
$$P(X_1, \dots, X_n) = \frac{1}{Z} \exp \left[ - \sum_{i=1}^k w_i f_i(\mathbf{D}_i) \right]$$

- Each feature  $f_i$  has a scope  $\mathbf{D}_i$
- Different features can have the same scope

Log-linear representation does not change the independence assertion made by the model

# Log-linear Markov networks

Example



$$P(x) = \frac{1}{Z} \exp \left( \sum_i w_i f_i(x) \right)$$

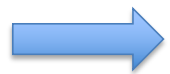
Weight of Feature  $i$       Feature  $i$

$$f_1(\text{Smoking}, \text{Cancer}) = \begin{cases} 1 & \text{if } \neg \text{Smoking} \vee \text{Cancer} \\ 0 & \text{otherwise} \end{cases}$$
$$w_1 = 1.5$$

# Log-linear Markov networks

- Features and weights can be reused for different factors
  - Typical: features designed by expert, weights learned from data.
- Log of the probability is linear in the weights
  - Ignoring Z, which is a constant for a given W

$$P(X_1, \dots, X_n) = \frac{1}{Z} \exp \left[ - \sum_{i=1}^k w_i f_i(\mathbf{D}_i) \right]$$


$$\ln[P(X_1, \dots, X_n)] \propto \sum_{i=1}^k w_i f_i(\mathbf{D}_i)$$

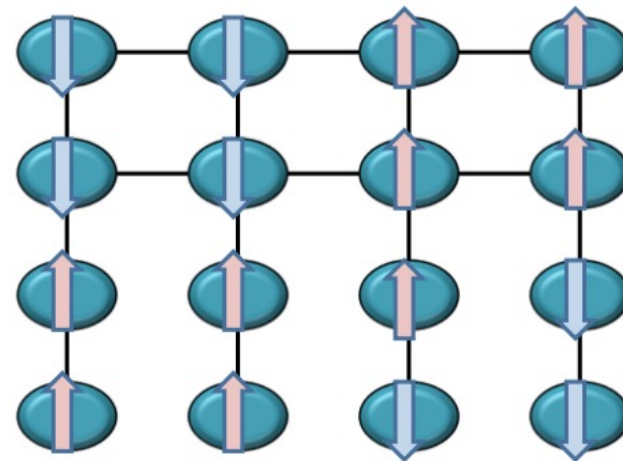
# Example: Ising Models

$$X_i \in \{+1, -1\}$$

$$P(X) \propto \prod_{i < j} w_{i,j} x_i x_j \prod_i u_i x_i$$

$$\phi_{i,j}(x_i, x_j) = w_{i,j} x_i x_j$$

$$\phi_i(x_i) = u_i x_i$$



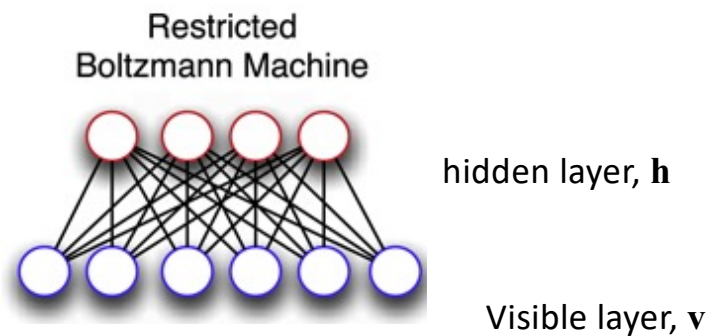
$$P(\xi) = \frac{1}{Z} \exp \left( - \sum_{i < j} w_{i,j} x_i x_j - \sum_i u_i x_i \right)$$

# Example: Boltzman Machine

$$P(\mathbf{v}, \mathbf{h}) = \frac{1}{Z} \exp(-E(\mathbf{v}, \mathbf{h}))$$

$$\begin{aligned} E(\mathbf{v}, \mathbf{h}) &= - \sum_{ij} v_i W_{ij} h_j - \sum_j b_j h_j - \sum_i c_i v_i \\ &= -\mathbf{v}^T W \mathbf{h} - \mathbf{b}^T \mathbf{h} - \mathbf{c}^T \mathbf{v} \end{aligned}$$

$$Z = \sum_{\mathbf{v} \in \{0,1\}^D} \sum_{\mathbf{h} \in \{0,1\}^K} \exp(-E(\mathbf{v}, \mathbf{h}))$$



# Special case of pairwise MRF: Metric MRFs

- Class of MRFs used for labeling (image segmentation, denoising)
- All  $x_i$  take values in label space  $V$
- We want  $x_i$  and  $x_j$  to take “similar” values (labels)
  - Smoothness constraint

- Energy function:

$$E(x_1, \dots, x_n) = \sum_i \epsilon_i(x_i) + \sum_{(i,j) \in \mathcal{E}} \epsilon_{i,j}(x_i, x_j)$$


Metric or distance function

$$P(x_1, \dots, x_n) \propto \exp(-E(x_1, \dots, x_n))$$

- Goal is minimizing the energy (higher the probability)

$$\arg \min_{x_1, \dots, x_n} E(x_1, \dots, x_n)$$

# Example: Metric MRFs

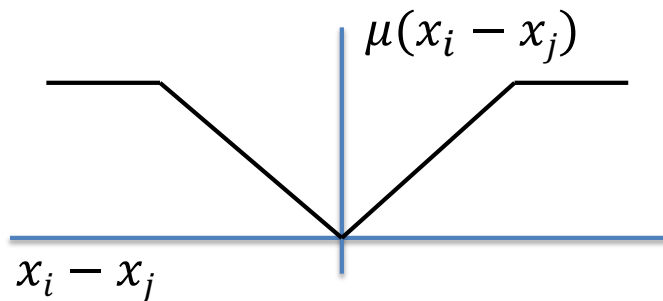
- Distance function on labels
  - Prefer neighboring nodes to have labels smaller distance apart
- Distance function  $\mu : V \times V \rightarrow \mathbb{R}$

– Example

$$\epsilon_{i,j}(v_k, v_l) = \mu(v_k, v_l)$$

Used in denoising and segmentation

$$\epsilon(x_i, x_j) = \min(c\|x_i - x_j\|_p, \text{dist}_{\max})$$



# Conditional random fields

- In practice we mostly care about distribution of  $\mathbf{Y}$  when  $\mathbf{X}$  is given not  $P(\mathbf{X}, \mathbf{Y})$
- A Markov network with directed dependencies
  - Observed variables  $\mathbf{X}$
  - Target variables  $\mathbf{Y}$
  - Undirected graph
    - $\{Y, X\}$  nodes
    - Set of factors  $\phi_1(\mathbf{D}_1), \dots, \phi_m(\mathbf{D}_m)$

$$P(\mathbf{Y}|\mathbf{X}) ?$$

# Conditional random fields

- Set of factors

$$\phi_1(\mathbf{D}_1), \dots, \phi_m(\mathbf{D}_m)$$

$$P(\mathbf{Y} \mid \mathbf{X}) = \frac{1}{Z(\mathbf{X})} \tilde{P}(\mathbf{Y}, \mathbf{X})$$

$$\tilde{P}(\mathbf{Y}, \mathbf{X}) = \prod_{i=1}^m \phi_i(\mathbf{D}_i)$$

$$Z(\mathbf{X}) = \sum_{\mathbf{Y}} \tilde{P}(\mathbf{Y}, \mathbf{X}).$$

Like Gibbs distribution, but normalized over  $\mathbf{Y}$  for given  $\mathbf{X}$

# Conditional random fields

- A CRF is parameterized the same as Gibbs distribution but normalized differently
- We don't need to model distribution over variables we do not care about (given variables)
- CRF allows models with highly expressive features without worrying about wrong dependency
- CRF has many applications specially in NLP and image segmentation

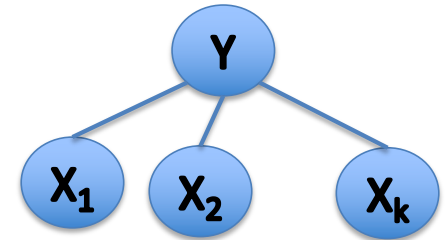
Why not using Naïve Bayes for image segmentation?

# Conditional random fields

- CRF is very useful when the feature variables are highly correlated
  - Text
  - Image
- Considers  $P(Y|X)$  not  $P(X, Y)$ 
  - Does not care about dependencies on  $X$
  - The dependency over  $X$  might be complex or poorly understood
  - Include continuous variables with no simple parametric forms
  - Can use domain knowledge to define features with no joint prob.

# CRFs and logistic model

Logistic regression is same as a conditional MN with a single query variable



- Binary valued  $X_1, \dots, X_k$  and  $Y$
- Pairwise potentials  $\phi_i(X_i, Y) = \exp \{w_i \mathbb{I}\{X_i = 1, Y = 1\}\}$

$$\phi_i(X_i, Y) = \begin{cases} \phi_i(X_i, Y = 1) = \exp\{w_i X_i\} \\ \phi_i(X_i, Y = 0) = 1 \end{cases}$$

- Single node potential  $\phi_0(Y) = \exp \{w_0 \mathbb{I}\{Y = 1\}\}$

$$\phi_0(Y) = \begin{cases} \phi_0(Y = 1) = \exp\{w_0\} \\ \phi_0(Y = 0) = 1 \end{cases}$$

- Conditional MN

$$P(\mathbf{Y} | \mathbf{X}) = \frac{1}{Z(\mathbf{X})} \tilde{P}(\mathbf{Y}, \mathbf{X}) \quad \text{where} \quad \tilde{P}(\mathbf{Y}, \mathbf{X}) = \prod_{i=1}^m \phi_i(D_i) \quad Z(\mathbf{X}) = \sum_{\mathbf{Y}} \tilde{P}(\mathbf{Y}, \mathbf{X}).$$

# CRFs and logistic model

- Binary valued  $X_1, \dots, X_k$  and  $Y$
- Pairwise potentials
- Single nodepotential

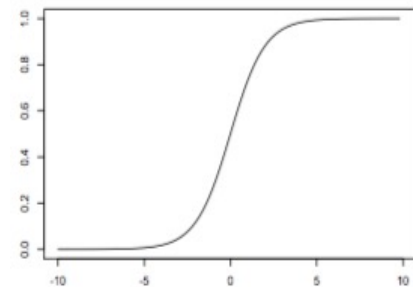
$$\phi_i(X_i, Y) = \exp \{w_i \mathbb{I}\{X_i = 1, Y = 1\}\}$$

$$\phi_0(Y) = \exp \{w_0 \mathbb{I}\{Y = 1\}\}$$

$$\tilde{P}(Y = 1 \mid x_1, \dots, x_k) = \exp \left\{ w_0 + \sum_{i=1}^k w_i x_i \right\}$$

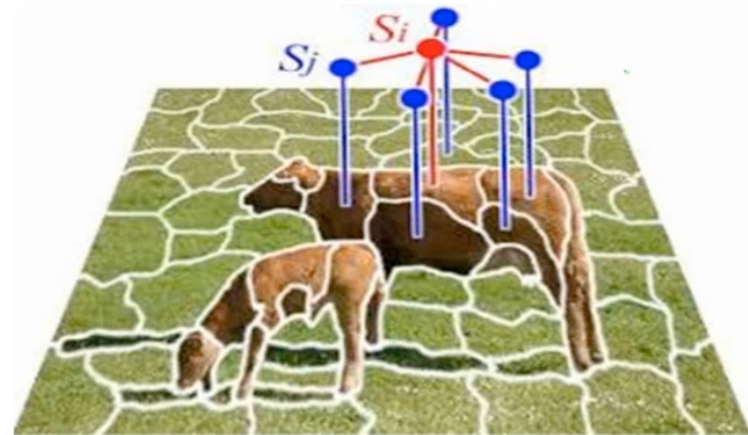
$$\tilde{P}(Y = 0 \mid x_1, \dots, x_k) = \exp \{0\} = 1.$$

$$P(Y = 1 \mid x_1, \dots, x_k) = \text{sigmoid} \left( w_0 + \sum_{i=1}^k w_i x_i \right)$$



# CRF for Image segmentation

- A node for each superpixel,  $X_i$
- Edge between nodes if regions are adjacent
- Class for each super pixel  $S_i$ 
  - Correlation between the pixel does not matter, when they are given
- Node factors can be any features of the image
  - Color histogram
  - Texture feature

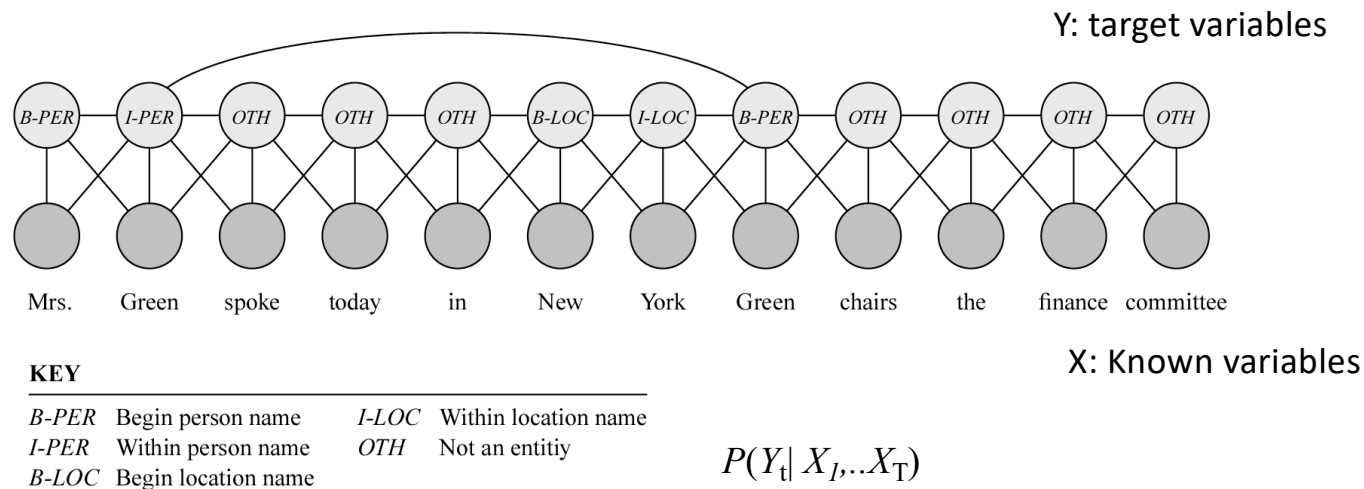


Factors for superpixel  $i$ :  $\Phi_{ij}(S_i, S_j, X_i)$ ,  $\Phi_i(S_i, X_i)$

# CRF for text analysis

## Speech labeling

- Identifying named entities
- Extracting structural information



Factors for word  $t$ :  $\Phi_{1_t}(Y_t, Y_{t+1})$ ,  $\Phi_{2_t}(Y_t, X_1, \dots, X_T)$

# Conditional Random Fields as Recurrent Neural Networks

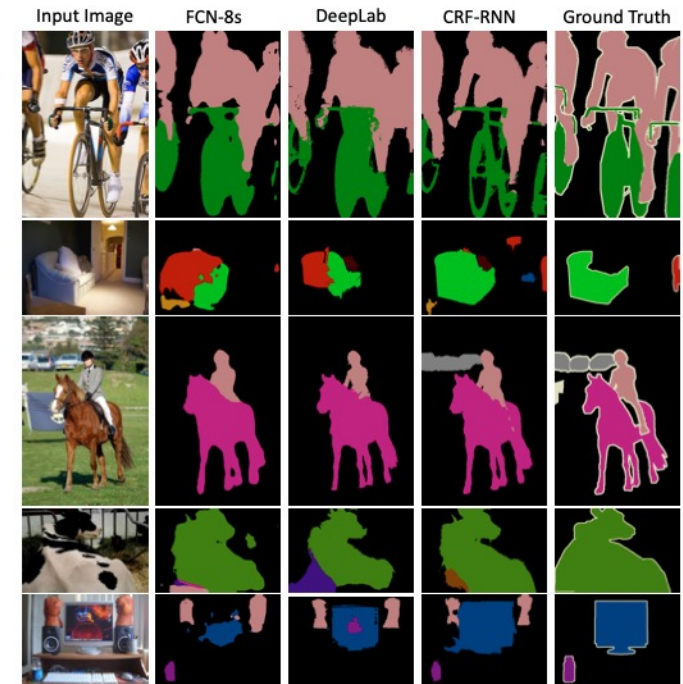
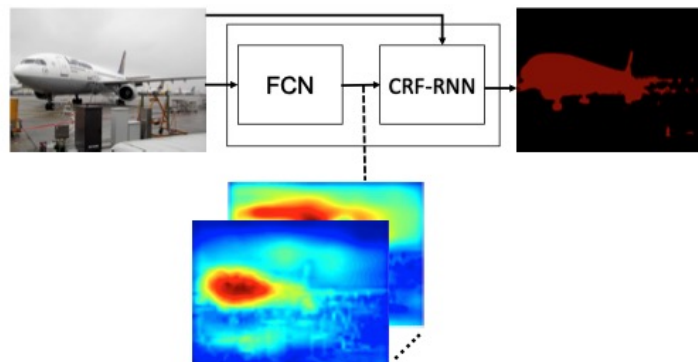
Shuai Zheng<sup>\*1</sup>, Sadeep Jayasumana<sup>\*1</sup>, Bernardino Romera-Paredes<sup>1</sup>, Vibhav Vineet<sup>†1,2</sup>, Zhizhong Su<sup>3</sup>,  
Dalong Du<sup>3</sup>, Chang Huang<sup>3</sup>, and Philip H. S. Torr<sup>1</sup>

<sup>1</sup>University of Oxford

<sup>2</sup>Stanford University

<sup>3</sup>Baidu Institute of Deep Learning

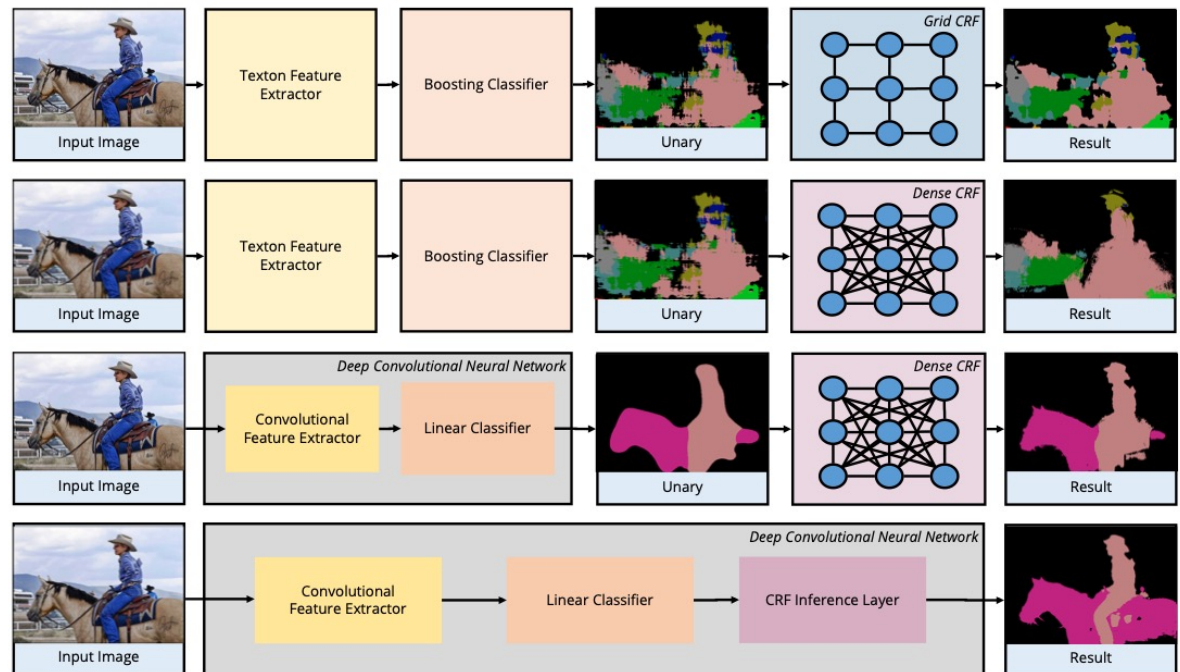
- The CNN produces a rough heatmap of class scores for every pixel in the image. These serve as the input energies to the CRF.
- The CRF links every pair of pixels in the entire image.



# Conditional Random Fields Meet Deep Neural Networks for Semantic Segmentation

Anurag Arnab\*, Shuai Zheng\*, Sadeep Jayasumana, Bernardino Romera-Paredes, Måns Larsson, Alexander Kirillov, Bogdan Savchynskyy, Carsten Rother, Fredrik Kahl and Philip Torr

- CRF as a Post-processing Step (Disconnected Pipelines)
- Deep Networks Informing CRF Potentials (Deep Feature Injection)
- Embedding Inference Within the Neural Network (End-to-End Joint Training)



Current state-of-the-art methods have all combined inference of a CRF within the deep neural network itself, allowing for end-to-end training

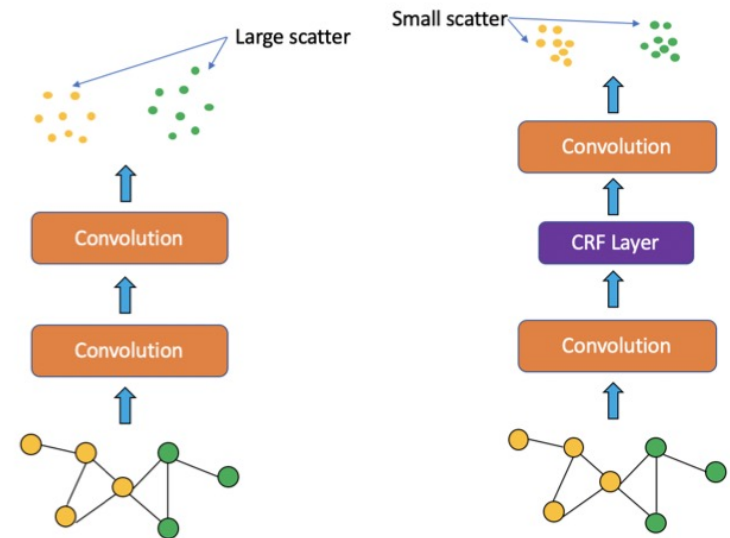
# CRF for graph convolutional neural network

Enhancing similarity in hidden features.

- Acts directly on the *hidden feature layers* of the network.
- The unary potential captures the original GCN node features, while the pairwise potential heavily penalizes connected or highly similar nodes if their hidden vectors drift apart.

$$P(H^{(l)}|B^{(l)}) = \frac{1}{Z(B^{(l)})} \exp(-E(H^{(l)}|B^{(l)})),$$

$$B_{i\cdot}^{(l)} = F_l(H_{i\cdot}^{(l-1)}, \hat{A}, W^{(l)})$$



(a) Standard Graph Convolutional Neural Network (b) Graph Convolutional Neural Network with CRF Layer

# BNs and MNs

## Directed

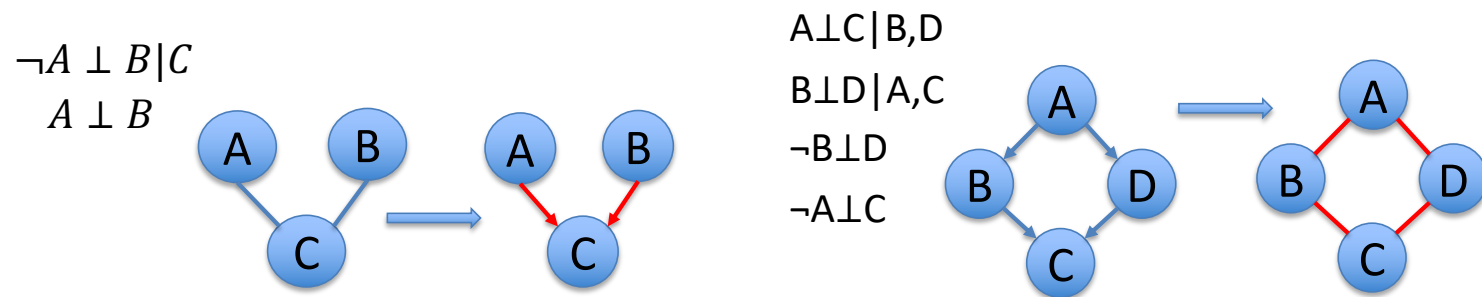
- Captures inter-causal reasoning, *eg* explaining a way
- Parameters interpretable, can be set by hand
- Usually easier parameter estimation
- Can easily generate data from the model
- Rich existing work in latent-variable models

## Undirected

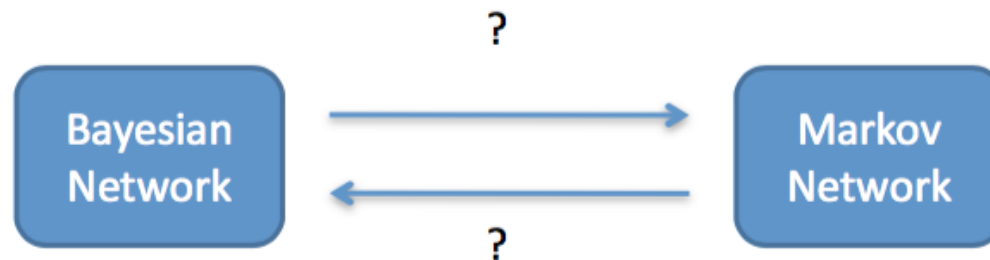
- Captures “affinity” Symmetrical. Cyclical graphs
- Param’s not so interpretable, usually learned from data
- Trickier parameter estimation, but not too bad
- Can easily add factors & overlapping features to the model
- Less work in latent-variable models, but there is some

# Transferring between BNs and MNs

- Some dependency cannot be captured by BN and some by MN



- How can we transfer one to the other one?

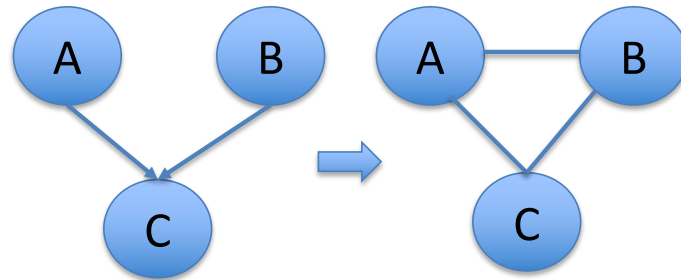
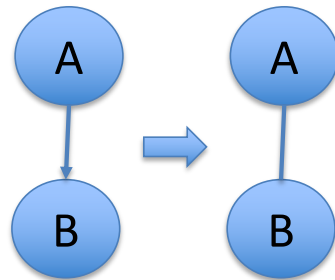


Can we capture all independencies?

## Bayesian Network to Gibbs Distribution

- Each conditional probability distribution is a factor
- Also works when conditioning on some evidence.
- Can we go from a Bayesian Network to an undirected graph that's an I-map?

# Bayesian Network to Gibbs Distribution



The Problem is on V structure  
→ moralization

It is an I-map but not a P-map

# Bayesian Network to Gibbs Distribution

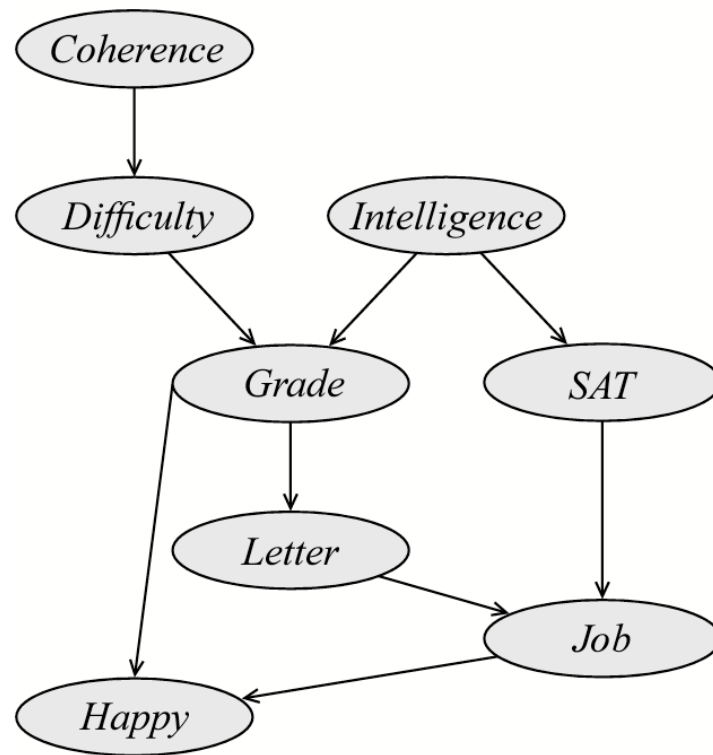
- In the Markov network, each factor must correspond to a subset of a clique.
- The “factors” in Bayesian networks are the CPDs.
  - Node + parents
- **Moralize** the graph: add an edge between any two nodes that share a child
- Moralizing ensures that a node and its parents form a clique.
  - But some independencies in the Bayesian network graph may be lost in the Markov network graph.

## BN Structure to MN Structure

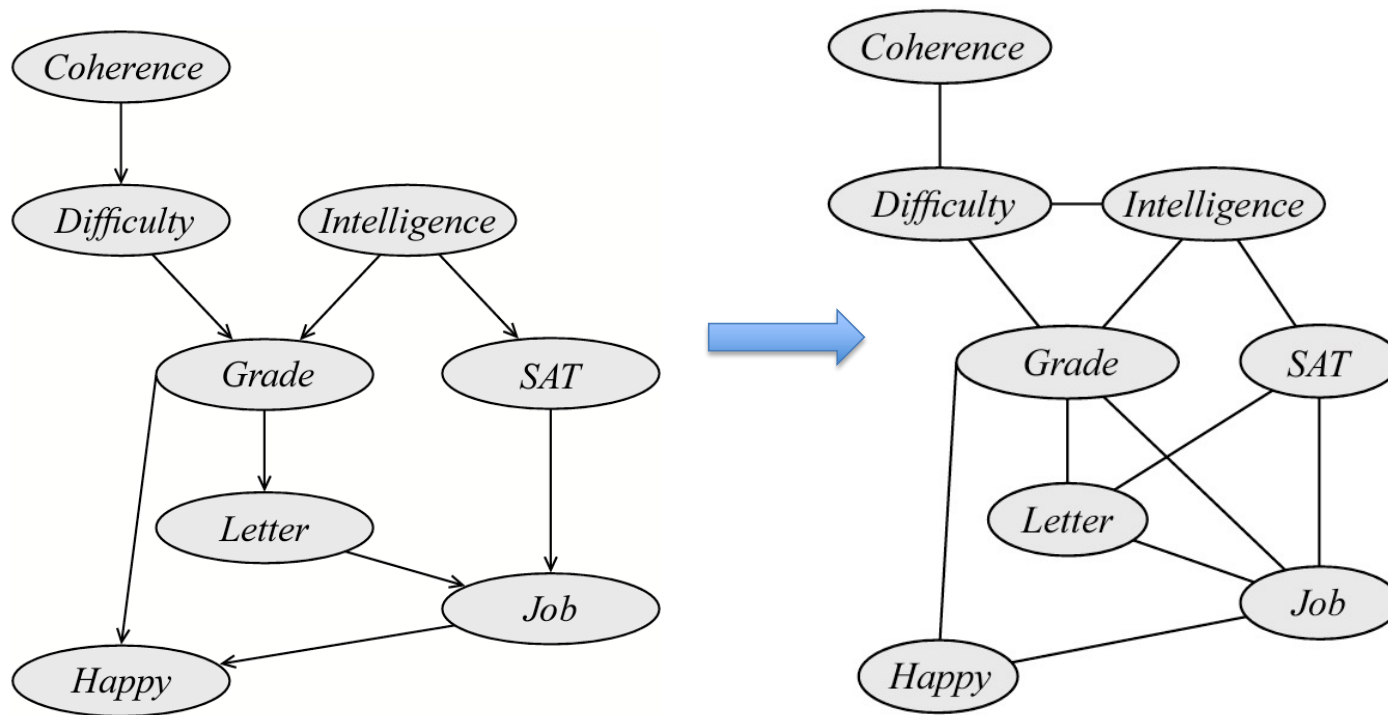
- Start with the Bayesian network skeleton of  $G$
- **Moralize** the graph: add an edge between any two nodes that share a child.

Result: moralized (undirected) graph is a minimal I-map for  $G$ .  
If  $G$  was moral already, P-map.

## Example of BN $\rightarrow$ MN



## Example of BN $\rightarrow$ MN



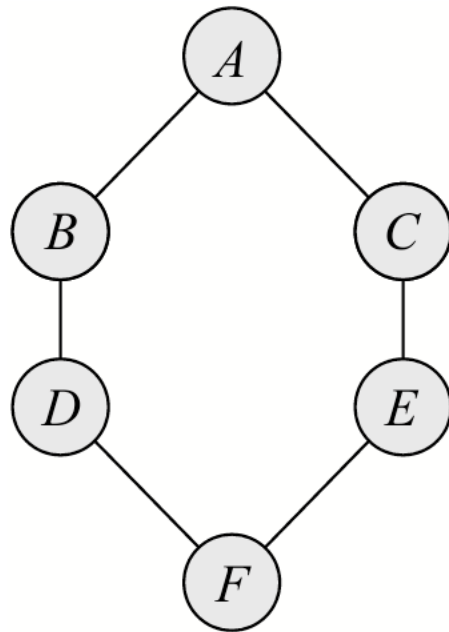
# Markov Network to Bayesian Network

## Building a Minimal I-Map

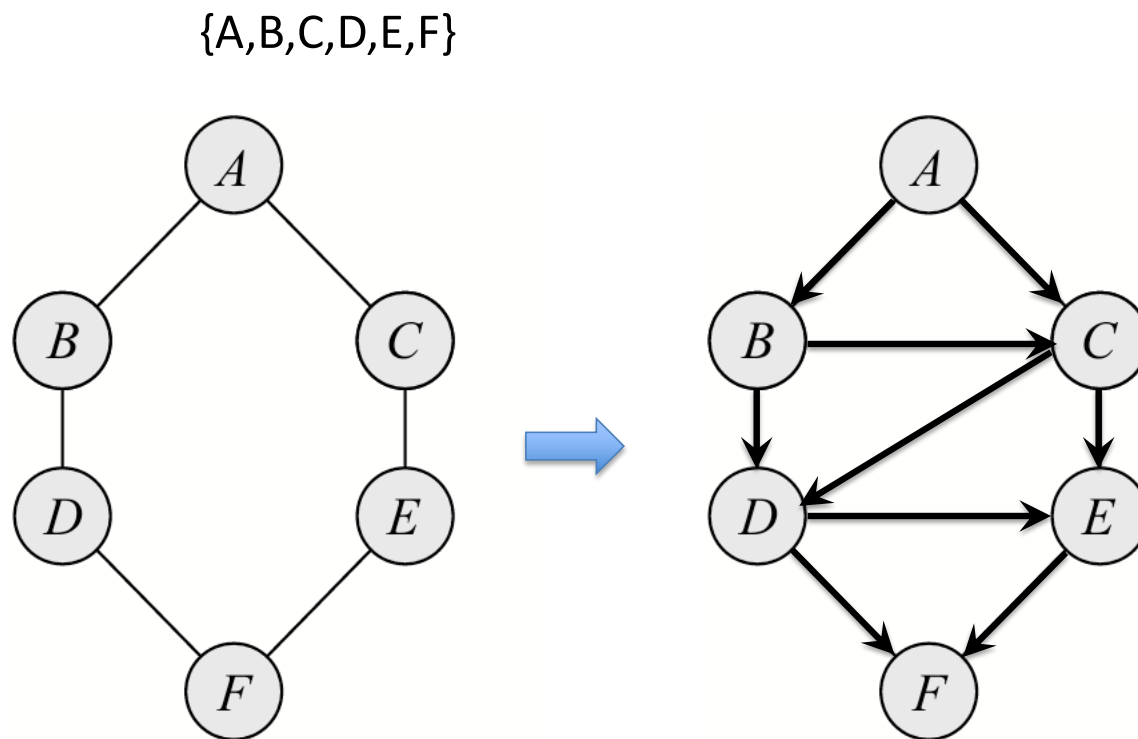
- Order variables arbitrarily, so that  $X_i$  precedes all its descendants.
- For  $i$  from 1 to  $n$ :
  - Add  $X_i$  to the network
  - Let **Parents**( $X_i$ ) be the minimal subset **S** of  $\{X_1, \dots, X_{i-1}\}$  such that  $X_i \perp (\{X_1, \dots, X_{i-1}\} \setminus \mathbf{S}) \mid \mathbf{S}$

# Example of $MN \rightarrow BN$

$\{A, B, C, D, E, F\}$

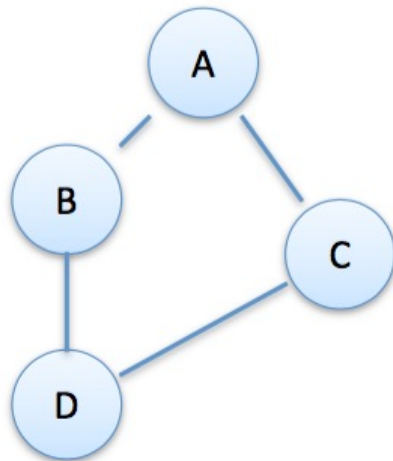


## Example of $MN \rightarrow BN$

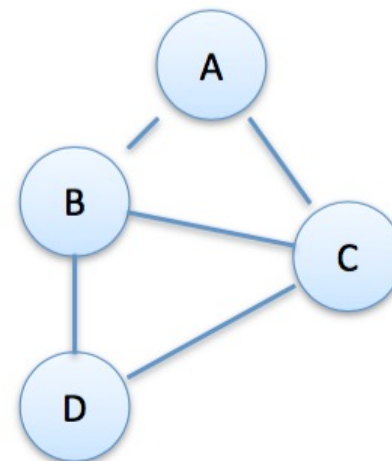


# Chordal Graphs

- Graph whose minimal cycles are not longer than 3.



Not Chordal Graph



Chordal Graph

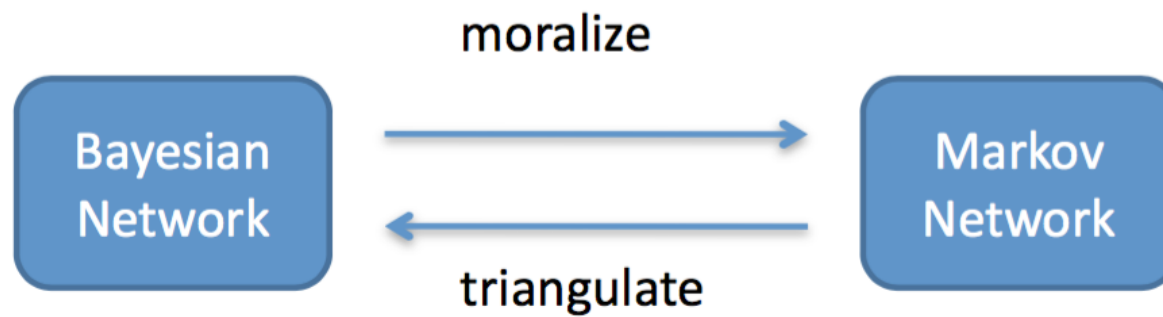
## Chordal graph for $MN \rightarrow BN$

- The Bayesian network we create cannot have any immoralities!
- If  $G$  is a minimal I-map Bayesian network for Markov network  $H$ , then  $G$  has no immoralities.
  - It is **chordal**,  
since any loop of length  $\geq 4$  in a Bayesian network graph must have immoralities.

# Markov Network to Bayesian Network

- Conversion from MN to BN requires triangulation
  - May lose some independence information.
  - May involve a lot of additional edges.
  - Different orderings of chain rule may yield different numbers of additional edges.

# Transferring between BNs and MNs

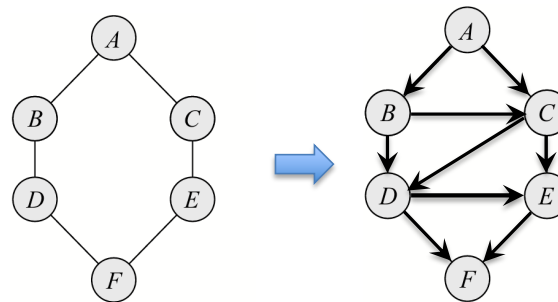


# Perfect transformation between BN and MN

- Bayesian network/Markov network conversion can lead to addition of edges and loss of independence information
- Is there a subset of distributions that can be captured *perfectly* in both models?
  - Yes! Undirected chordal graphs.

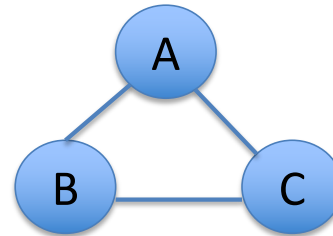
# Theorem

- If  $H$  (a Markov network) is non-chordal, then there is no Bayesian network  $G$  such that  $I(G) = I(H)$ , i.e., no P-map.
- Why?
  - Minimal I-map for  $G$  must be chordal. If  $G$  is an I-map for  $H$ , it must include some additional edges not in  $H$ , but that eliminates independence assumptions. So  $I(H)$  can't be perfectly encoded.



# Factor Graph

- Graphical models
  - Bayesian Network
  - Markov Network
  - Factor Graph



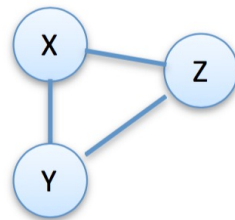
- Factor graph is for representation of factorization of a distribution
  - Alternative representation for parameterization of a Markov Network
  - typically equivalent to independence statements amongst (sets of) variables in the distribution.

# Representing factorization in Graphs

- Belief Network, Bayesian Network
  - Each factor is a conditional distribution
  - Generative models, AI, statistics
  - Corresponds to a DAG
- Markov Network
  - Each factor corresponds to a potential (non negative function)
  - Related to the strength of relationship between variables but not directly related to dependence
  - Useful for collective phenomena such as image processing
  - Corresponds to an undirected graph
- Factor Graph
  - A barebones representation of the factorization of a distribution
  - Often used for efficient computation and deriving message passing algorithms.

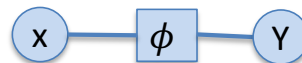
# Factor graph

- Makes the structure of the Gibbs distribution explicit

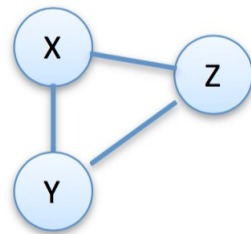


Which Gibbs dis.?

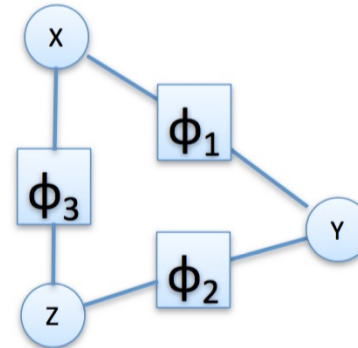
- Contains two types of node
  - Random variables
  - Factors over the variables
- Contains edges between variable nodes and factor nodes.
- Each factor node  $V_\phi$  is associated with one factor  $\phi$ , which its scope is neighboring variables of  $V_\phi$



# Factor graph



pairwise Markov network



all cliques

